Thermodynamics and Statistical Physics

3rd Exercise Class

Heat and Internal energy; 1st Law of Thermodynamics; Heat Capacity, Specific Heat and Latent Heat; Ideal and van der Waals Gases

Exercises for the Class

1. A gas is contained in a cylinder fitted with a frictionless piston and is taken from state a to state b along the path acb shown in Figure 1. 80 J of heat flows into the system, and the system does 30 J of work.

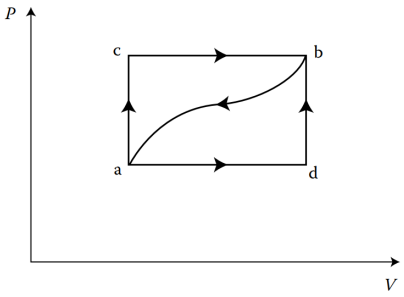


Fig. 1

   (a) If instead the work done by the gas system is only 10 J along adb, how much heat flows into the system?

   (b) When the system is returned from b to a along the curved path, the work done on the system is 20 J. What is the heat transfer?

   (c) If $U_a = 0$ and $U_d = 40$ J, find the heat absorbed in the processes ad and db.

2. 2.5 moles of oxygen is initially at $T = 300$ K and $p = 1.0$ atm. The gas is then taken on a three-step cycle: (i) The pressure and volume increase in such a way that p is proportional to V, until $p = 2.0$ atm; (ii) pressure is reduced at constant volume to 1.0 atm; and (iii) volume is reduced at constant pressure until the initial state is reached.

   (a) Find the internal energy and volume occupied by the gas in its initial state.

   (b) Find ΔU, W, and Q for each step in the process.

   (c) Find the net values of ΔU, W, and Q for the entire cycle.

   (d) Explain why the signs on your answers in (c) make sense. Note that for diatomic ideal gases $\gamma = 7/5$.

3. One mole of an ideal gas goes through a process described by the equation $p(V) = p_0 + \alpha/V$, where p_0 and α are constants.

   (a) Determine the molar specific heat during the process as a function of the volume.

   (b) The gas volume changes from V_1 to V_2. Determine the change of its internal energy, the work done on the gas and the absorbed heat.

4. The state of equation for van der Waals gases reads as

$$\left(p + \frac{a}{v^2}\right)(V - nb) = nRT,$$

   where a and b are constants and $v = V/n$ is the molar volume. Calculate the work done by a van der Waals gas in expanding from a volume V_1 to a volume V_2:

   (a) at constant pressure p;

   (b) at constant temperature T.

5. A 2.5 kg piece of hot lead at 95 °C is dropped from a 35 m high tower into 10 liters of water at 20 °C.

   Compute separately the temperature rise in the water (once equilibrium is reached) due to the lead's (a) thermal properties and (b) gravitational energy. Compare and assess the results in (a) and (b).

   The specific heat of water is 4200 J/(kg K) and the specific heat of lead is 130 J/(kg K).

6. The molar specific heat of many materials at low temperatures is found to obey the Debye law $c_V = A(T/\theta)^3$ where A is a constant equal to $2.68 \cdot 10^3$ J/(K mol) and with the Debye temperature θ taking different values for different materials. For diamond it is 2230 K.

   (a) How much heat is required to raise one mole of diamond from 20 K to 100 K?

   (b) What is the average molar specific heat in this range?

7. On a hot day you want to cool 0.5 liter of tap water initially at 25 °C by adding ice.

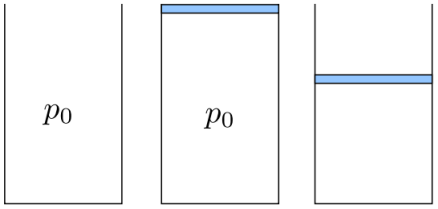
   (a) If you add 45 g of ice, what is the final equilibrium temperature of the mixture?

(b) How much ice would you need to add to the original sample of water (0.5 liter at 25 °C) so that the last of the ice melts as the mixture reaches 0 °C?

The specific heat of ice is $c = 2050\text{J}/(\text{kg K})$, specific heat of water is 4200 J/(kg K), specific latent heat of fusion is 334 kJ/kg.

Homework to Be Submitted

1. There is air in a vertically oriented, thermally insulated cylinder that is 2 dm high and has a base area of 1 dm^2. A thermally insulated piston of mass 10 kg is carefully placed on top of the cylinder. The piston seals the air in the cylinder airtight. The movement of the piston is impeded by slight friction.



Measured from the top of the cylinder, at what distance will the piston come to rest? The external air pressure is 10^5 Pa.

Homework for Practice

1. Determine the equation of the adiabatic process for a van der Waals gas.

2. What amount of heat is to be transferred to nitrogen in the isobaric heating process for that gas to perform the work 2.0 J?

3. One mole of an ideal gas with heat capacity at constant pressure C_p undergoes the process $T = T_0 + \alpha V$, where T_0 and α are constants. Find:

   (a) heat capacity of the gas as a function of its volume;

   (b) the amount of heat transferred to the gas, if its volume increased from V_1 to V_2.

4. A sample of water of mass $m_1 = 300$ g and of temperature $T_1 = 20$ °C is in a thermos flask, and warmer water of mass $m_2 = 600$ g and of temperature $T_1 = 80$ °C is poured to it. After the thermal equilibrium is reached the temperature is measured. Next time, because we are curious, we swap the two samples of water, so the colder water of mass m_1 and of temperature T_1 is poured into the thermos in which there is the other sample of water of mass m_2 and of temperature T_2. Surprisingly it is found that the equilibrium temperature in the second case is different from the one measured in the first time.

   (a) How can you explain this difference?

   (b) The difference between the two equilibrium temperatures is 1.9 °C. What can you calculate from this?